

Nicholas D. Gray, Ph.D.

QUANTITATIVE RESEARCHER

ndgray00@gmail.com | 513-504-0179 | Maryland Heights, MO | ndg13b.github.io/Resume | github.com/ndg13b

Researcher with a Ph.D. in Cognitive Psychology and nine years designing, running, and analyzing behavioral health studies, including large-scale clinical trials on cognitive aging. Works across mixed methods — instrument development, survey fielding, qualitative analysis, and quantitative analysis from descriptives through structural equation modeling — and brings four years of industry data engineering in SQL and Python to data preparation and reporting. Published in peer-reviewed health and aging journals.

RESEARCH & ANALYTIC SKILLS

Research Methods	Experimental design, Survey & instrument development, Sampling & power analysis, IRB preparation & submission, Qualitative interviews & analysis, Mixed-methods integration
Statistical Analysis	Descriptive & inferential statistics, ANOVA, Regression (linear & logistic), Multivariate analysis, Structural equation modeling, Meta-analysis
Tools	Python (proficient), R, SPSS, SQL / Microsoft SQL Server, Qualtrics, MATLAB, Microsoft Excel (proficient)
Data & Reporting	Data cleaning & merging, ETL, Automated reporting pipelines, Data visualization
Communication	Manuscript writing, Stakeholder reporting, Technical writing, Presentation & teaching

PROFESSIONAL EXPERIENCE

Data Scientist

Oct 2022 – Present

Safety National Casualty Corporation · St. Louis, MO

- Design custom analyses out of exploratory investigation, tailoring method and reporting to the question each stakeholder is actually asking.
- Extract and join data from large, complex sources in Microsoft SQL Server, then clean, merge, and reshape it in Python for analysis.
- Built an automated reporting pipeline that calls an LLM to generate written insights alongside the quantitative results.
- Build analytics reports and dashboards for internal teams and external stakeholders, and validate output from established predictive models.

Postdoctoral Researcher — Quantitative Analysis

Aug 2019 – Oct 2022

Institute for Successful Longevity, Florida State University · Tallahassee, FL

- Led statistical analysis for large-scale behavioral clinical trials on healthy cognitive aging, working across longitudinal datasets and multiple analysis platforms.
- Designed and programmed studies end to end, preparing and submitting the IRB applications and building and fielding survey instruments in Qualtrics.
- Established statistical performance norms and built regression models identifying the variables that drive cognitive outcomes.
- Integrated qualitative data from surveys and interviews with quantitative measures to produce findings neither stream would support alone.
- Published results in peer-reviewed journals and presented findings at national conferences.

Graduate Researcher

Aug 2013 – May 2019

Department of Psychology, Florida State University · Tallahassee, FL

- Modeled experimental data with logistic regression and related statistical methods across several software platforms.
- Designed, programmed, and ran grant-funded experimental research; managed the lab, recruiting and supervising undergraduate researchers.
- Taught laboratory sections of Cognitive Psychology for three years and co-instructed an online Careers in Psychology course, translating research concepts and methods for non-expert audiences.

EDUCATION

Ph.D., Cognitive Psychology

2019

Florida State University

Thesis: The causes of and influences on being reminded

M.S., Cognitive Psychology

2016

Florida State University

Thesis: Avoiding interference: Differentiation and reminding

B.S., Psychology, cum laude

2012

Ohio State University

Certification: Machine Learning Specialization — DeepLearning.AI & Stanford Online (Coursera), 2026. Supervised learning (regression, classification, neural networks, decision trees), unsupervised learning, recommender systems, reinforcement learning

Graduate methods training: ANOVA, regression, meta-analysis, structural equation modeling

Additional coursework: Machine Learning with Python; Microsoft SQL Server (independently completed)

SELECTED PUBLICATIONS

Gray, N., & Charness, N. (2022). Technology obsolescence across the adult lifespan in a USA internet sample. *Frontiers in Public Health*, 10, 1005822.

Gray, N., Yoon, J.-S., Charness, N., Boot, W. R., et al. (2021). Relative effectiveness of general versus specific cognitive training for aging adults. *Psychology and Aging*, 37(2), 210–221.

Carr, D. C., Tian, S., He, Z., et al., incl. Gray, N. (2022). Motivation to engage in aging research: Are there typologies and predictors? *The Gerontologist*, gnac035.

Charness, N., Boot, W. R., & Gray, N. (2020). Mobile monitoring and intervention technology (MMI) for adaptive aging. In *Mobile Technology for Adaptive Aging: Proceedings of a Workshop* (pp. 21–40). National Academies Press.

Conference presentations at the Psychonomic Society (2018) and the Cognitive Aging Conference (2022).

SELECTED PROJECTS

adherence – Python · github.com/ndg13b/adherence

Scores how regular a behavior is — not just how often it happens — from nothing but a list of timestamps, then forecasts when the next session will land. The same model applies to medication timing, exercise, physiotherapy, or a training program.

CivicGateway – JavaScript · civicgateway.org

A nonpartisan site showing upcoming elections and elected officials by city; static front end over a Supabase/PostgreSQL back end with a schema designed from scratch, row-level security, and an offline fallback.

tuned-in – JavaScript · ndg13b.github.io/tuned-in

A teaching collection pairing pop-culture clips with the cognitive psychology they demonstrate — and, just as often, get wrong. Each concept is labeled accurate, overstated, or myth.